



Department of Electronics and Communication Engineering
Academic Year 2021 – 2022 (Even Semester)

Degree, Semester & Branch : VI Semester B.E. ECE A&B

Course Code & Title : CS8792 Cryptography and Network Security

Name of the Faculty member : Mr.A.Rameshbabu

Innovative Practice Description

- **Unit / Topic:** Unit V / Malicious software
- **Course Outcome:** CO 5
- **Topic Learning Outcome:** TLO 19
- **Activity Chosen:** Flipped Classroom
- **Justification:**
 - A flipped classroom is a type of blended learning where students are introduced to content at home and practice working through it at College.
- **Time Allotted for the Activity:** 50 Minutes
- **Details of the Implementation:**
 - This is the reverse of the more common practice of introducing new content at college, then assigning homework and projects to complete by the students independently at home.
 - In this blended learning approach, face-to-face interaction is mixed with independent study—usually via technology. In a common Flipped Classroom scenario, students might watch pre-recorded videos at home, then come to school to do the homework armed with questions and at least some background knowledge.
 - The concept behind the flipped classroom is to rethink when students have access to the resources they need most. If the problem is that students need help doing the work rather than being introduced to the new thinking behind the work, then the solution the flipped classroom takes is to reverse that pattern.

• **CO – PO / PSO mapping:**

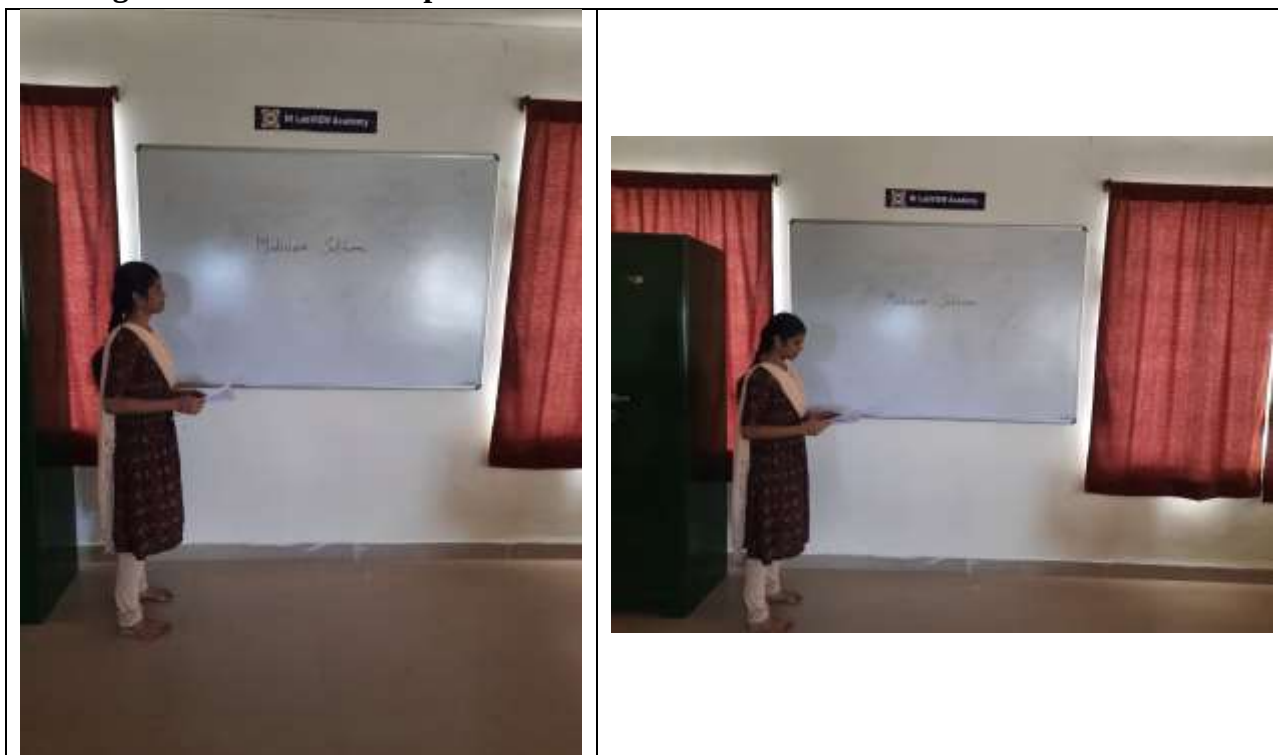
CO	PO 1	PO 2	PO 3	PO 4	PO 8	PO 9	PO 10	PO 12	PSO 2
CO3	2	2	2	1	1	1	1	1	2

(1 – Low 2 – Moderate 3 – High)

• **PO / PSO mapped:**

Innovative practice	PO 9	PO 10	PSO 2
Justification for correlation	Students will function effectively as an individual, and as a member or leader in Flipped Classroom activity to discuss about various Malicious software, So Course outcome is mapped at level 1	In Flipped Classroom activity students will make effective presentations on Malicious software and its classifications, hence Course outcome is mapped at level 1	Knowledge of Malicious software's is used to become a Excel in latest technologies of Embedded Systems, So Course outcome is mapped at level 2

• **Images / Screenshot of the practice:**



• **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

- Students eagerly participated and enjoyed this activity.

❖ **Benefit of the practice:**

- Outcome attainment would have increased due to innovative practice over conventional practice.
- Students easily understand the concepts of Malicious software's
- Students can answer easily about the classifications of Malicious software's

❖ ***Challenges faced in implementation:***

- Faced issues while make the students to understand about Flipped Class room Activity rules.

References:

- ❖ <https://www.geeksforgeeks.org/malwares-malicious-software/>
- ❖ <https://www.myrasecurity.com/en/what-is-malware/>
- ❖ <https://pchtechnologies.com/what-is-malicious-software/>
- ❖ <https://purplesec.us/common-malware-types/>



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Course Code & Title : CS8792 Cryptography and Network Security

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Innovative Practice Description

- **Unit / Topic:** Unit IV / Challenge Response protocols - Authentication applications
- **Course Outcome:** CO 4
- **Topic Learning Outcome:** TLO 15
- **Activity Chosen:** jigsaw
- **Justification:**
 - The concept of Authentication protocols & Applications was very important for Cryptography and Network Security. The activity 'JIGSAW' is used to understand the Authentication concepts and enhance the learning through this collaboration.
 - The activity 'JIGSAW' is used to recall the concepts learnt and enhances the learning through this collaboration.
- **Time Allotted for the Activity:** 50 Minutes
- **Details of the Implementation:**
 - The material related to the topic, will be posted in CANVAS one week before the conduct of activity.
 - Activity: JIGSAW - The Class Strength is divided into 6 Groups.
 - Each group is allotted with a colour codes – **RED, GREEN, BLUE, BROWN, PINK, VIOLET.**
 - Each Expert group is allotted with one topic
 - ✓ Message Authentication
 - ✓ Digital signature
 - ✓ Authentication protocols
 - ✓ Authentication applications
 - ✓ Kerberos
 - ✓ X.509 protocol
 - The expert group members discuss the topic in detail for 20 minutes with the learning materials.
 - The Expert group members should go to their original colour group and discuss the points (30 Minutes) to the other members.

- All the members in home group learnt about all the topics through expert group members.
- Finally one from each colour group should summarize the points of the topic Authentication protocols & applications to the class (20 Minutes).
- Review of points and conclusion of the activity (5 minutes).
- Thus the concept of Authentication protocols & applications is known to all by involving the students actively.

• **Group wise details:**

S.No.	Group Name	Topic Details
1.	Code - RED	Message Authentication
2.	Code - GREEN	Digital signature
3.	Code - BLUE	Authentication protocols
4.	Code - BROWN	Authentication applications
5.	Code - PINK	Kerberos
6.	Code - VIOLET	X.509 protocol

• **CO – PO / PSO mapping:**

CO	PO 1	PO 2	PO 3	PO 4	PO 8	PO 9	PO 10	PO 12	PSO 1
CO3	2	3	3	2	1	1	1	1	1

(1 – Low 2 – Moderate 3 – High)

• **PO / PSO mapped:**

Innovative practice	PO 9	PO 10	PSO 1
Justification for correlation	Students will function effectively as an individual, and as a member or leader in Jigsaw activity to discuss about various Authentication applications, So Course outcome is mapped at level 1	In Jigsaw activity students will make effective presentations on the Challenge Response protocols, hence Course outcome is mapped at level 1	Knowledge of Challenge Response protocols, topic used to innovate new ideas to solve existing/novel problems., So Course outcome is mapped at level 1

- **Images / Screenshot of the practice:**



- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

- Students eagerly participated and enjoyed this activity.

- ❖ **Benefit of the practice:**

- Students easily understand the concepts of Challenge Response protocols & Authentication applications
- Students can answer easily about different Authentication applications like Kerberos & X.509 protocol

- ❖ **Challenges faced in implementation:**

- Lack of preparation: Making the students to learn the materials is the challenging task.
- Each student has to read the material posted in the course website.
- Non participation in the activity: All the students should be made involve in the activity
- Time management: The Jigsaw activity should be able to complete in the planned duration.

References:

- ❖ <https://www.jigsaw.org/>
- ❖ https://www.newcastle.edu.au/data/assets/pdf_file/0016/109600/Jigsaw-learning-activity.pdf
- ❖ William Stallings, Cryptography and Network Security, 6th Edition, Pearson Education, March 2013.
- ❖ NPTEL Video Course on “Message Authentication Code” Prof. D.Mukhopadhyay, Department of Computer Science and Engineering, IIT Kharagpur.